Ch-02
Monte CARLO METHOD.

<u>Monte carlo method:</u> The Monte-carlo method is way to solve problems by using random examples to guess the chances of different results.

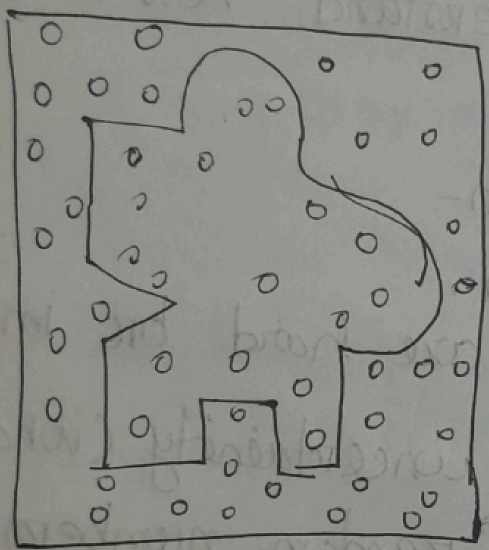
→ Is a way to simulate real-life uncertainity using random numbers to understand how a system behaves.

→ Monte-carlo helps when-
- Things are random.
- Exact formulas are hard or impossible.

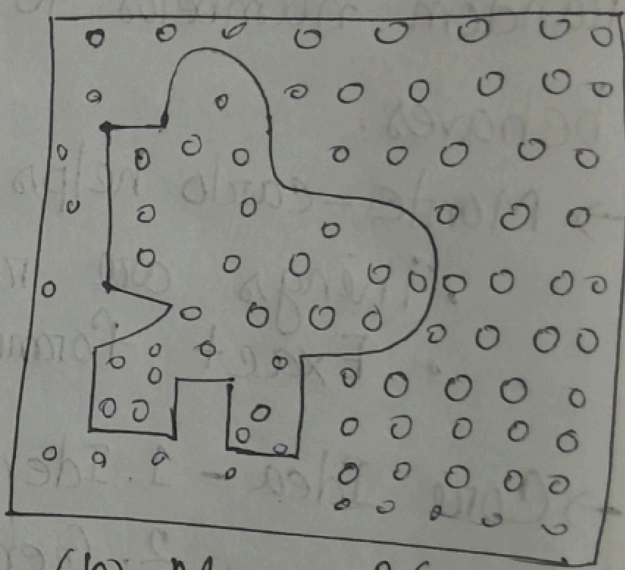
→ Core Idea - 1. Identify uncertainity (what is random)
2. Generate random numbers.
3. Repeat many times & analyze results.

(Example-2.1): Calculate the Area of an Irregular Figure using Monte carlo method

S: Enclose the Irregular figure in a known regular shape.



(a) $M/N = \frac{17}{50}$



(b) $M/N = \frac{26}{84}$.

N = Total number of random points generated inside the square

M = no. of points lying inside the irregular figure.

Random points are generated using random numbers between 0 & 1.

→ The square side is 9cm. the coordinate of each random points are calculated as.

$$x = r_x \times 9$$

$$y = r_y \times 9.$$

r_x & r_y are random numbers between 0 & 1.

Let, random numbers: 0.96 0.42 0.71 0.15 0.0

0.30

Point-1. $x_1 = 0.96 \times 9 = 8.64$

$y_1 = 0.42 \times 9 = 3.78$

Point 2, $x_2 = 0.71 \times 9 = 6.39$

$y_2 = 0.15 \times 9 = 0.99$

Point-3 $x_3 = 0.05 \times 9 = 0.45.$

$y_3 = 0.30 \times 9 = 2.7$

Each point is plotted inside the square.

Each generated point is examined:
- If the point in lies inside the irregular shape, it is counted.
- If it lies outside, it is ignored

Monte carlo area relation.

$$\frac{\text{Area of irregular figure}}{\text{Are of square}} \approx \frac{M}{N}.$$

$$\frac{F}{A} = \frac{M}{N} \Rightarrow F = \frac{M}{N} \times A$$

for fig-a, Area ratio $= \frac{M}{N} = \frac{17}{50} = 0.34$

it's far from True value =

Given, True value $= 0.3126$.

for fig-b $\Rightarrow$ Area ratio $= \frac{M}{N} = \frac{26}{84} = 0.3095$

which is closer. $= 0.3095$

$A = 9 \times 9 = 81\ cm^2$

$\therefore F = 0.3095 \times 81$

$= 25.07\, cm^2$.

~~Error~~, Larger the number of points, more accurate will be the result. mean,

As monte carlo depends on random number to generate results so smaller points-এ random updown effect বেশি থাকে। If we take larger points হলে average-এ কমে যায় এবং true value-র কাছাকাছি থাকে।

Error :- $\quad \text{Error} \propto \frac{1}{\sqrt{N}}$. Error decrease slowly Doubling N does not halve error.

$$\Rightarrow \frac{1}{\sqrt{N_{new}}} = \frac{1}{2} \times \frac{1}{\sqrt{N_{old}}}$$

$$\Rightarrow \sqrt{N_{new}} = 2\sqrt{N_{old}}.$$

$$N_{new} = 4 N_{old}.$$

→ mean, If we want to Error half we have to increase N 4 times.

→ To gain one additional correct decimal digit, N must be increased 100 times.

Then slow that Monte carlo method converge slowly.

(Example-2.2-A) Gambling Game -

Let us considered a game in which an unbiased coin is repeatedly flipped. For each flip you have to pay RS.1. when the difference between heads tossed & tails tossed becomes 3, you get Rs. 8.. If the required difference is obtained in less than 8 flips. you win some money & If it is in more than 8 flips you lose.

when a coin is flipped, the probability of head or tail is same that is 50%, but there are 10 possible one digit random numbers.

Let half of these (0,1,2,3,4) can be taken for Head, & the remaining half (5,6,7,8,9) for the Tail.

Let simulate some game in table.

Game No.	SL No	RN	Head or Tail	Cumulative Head	Cumulative tail	Difference.
1	1	5	T	0	1	1.
	2	9	T	0	2	2.
	3	3	H	1	~~0~~2	1.
	4	6	T	1	3	2.
	5	4	H	2	3	1
	6	8	T	2	4	2
	7	6	T	2	5	3.
Game No - 2	1	8	T	0	1	Win Re. 1 1
	2	1	H	1	1	0
	3	6	T	1	2	1.
	4	4	H	2	2	0
	5	3	H	3	2	1.
	6	7	T	3	3	0
	7	8	T	3	4	1.
	8	5	T	3	5	2
	9	9	H	~~6~~3	~~5~~6	~~1~~3

Loss Re 1

In the first game - coin flipped = 7.

cost = Re - 1.

Total cost = 7 × 1 = Rs. 7.

win = Rs. 8.

get = 8 - 7 = Rs - 1. gain

But monte carlo says → খেলায় না খেলে

let simulate so many times.

→ so in 7 game: let's.

Game	flipped coin to win/Loss
1	7
2	9
3	5
4	11
5	11
6	9
7	7

Total cost → Total flip = 7+9+5+11+9+7
= 59.

∴ total cost = 59 × 1 = 59.

Total reward :

Per game $= Rs.\ 8.$

total $u = 7.$

$= 8 \times 7 = Rs. 56.$ (Total reward).

$\therefore$ Win / Loss $=$ ~~59 56~~ $= 56 - 59$

$= -3.$

$\therefore$ Los Rs. 3 .

→ Average → $\frac{59}{7} = 8.43.$

$\therefore$ So per game total flip need 8.43.

total cost $=$ Rs. 8.43

But reward $=$ Rs. 8

Average loss $= 8 - 8.43 = 0.43.$

→ If the simulation is continued over a sufficiently long period that is for a large number of trials, the estimate close to the true mean can be obtained. which in this case 9 flips for each game. cost = Rs. 9. Rew = Rs. 8.

Average loss = Re. 1 per game

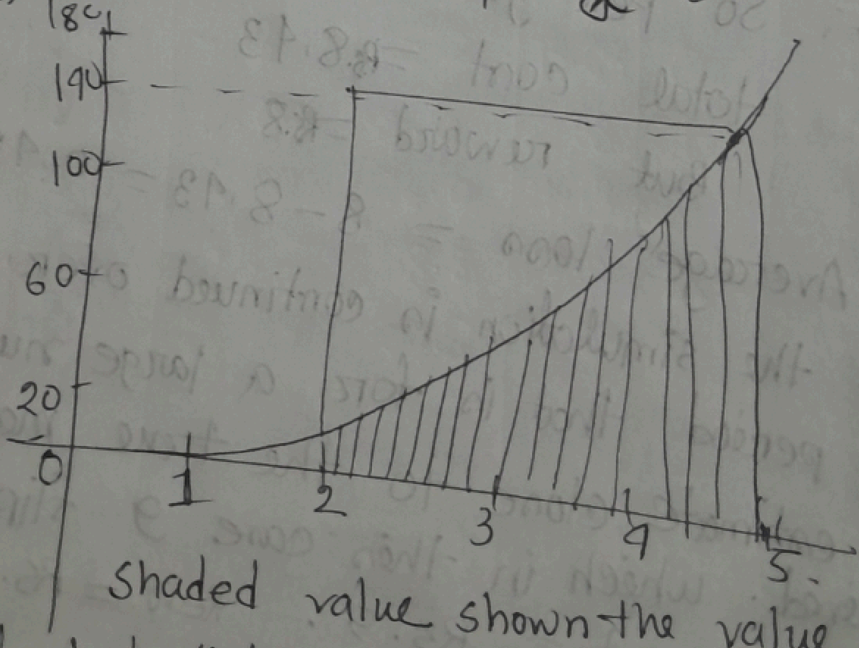
(Example-2.3) let consider a single variable function $I = \int_2^5 x^3 dx$

$= \int_a^b f(x)\, dx$.

$$I = \left[\frac{x^4}{4}\right]_2^5$$

$$= \frac{5^4 - 2^4}{4} = \frac{625-16}{4} = 152.25$$

$f(x) = x^3$ · $x=2$, $x=5$.



Shaded value shown the value of the integral. Let this area be enclosed in a rectangle ABCD whose area is known.

$(140 \times 3) = 420 = A$

width $= 5-2 = 3$.

$y = x^3 = 5^3 = 125$.

Height $= 140$.

The ratio of the shaded area I to the area A of rectangle ABCD is M to N.

$$I = \frac{M}{N} \cdot A$$

(Larger the value of N more accurate the value of result).

To mark a random point we need two random numbers to generate the two co-ordinate x & y.

Let us take two digit random number between 20 & 50 to represent the x co-ordinate.

→ random $N = 20$ to 50 But we need $x = 2$ to 5.

So $\frac{5-2}{50-20} = \frac{3}{30} = 0.1$. So, $\left[x = 0.1 r \quad r = \text{random number} \right]$

→ To generate y co-ordinate take random number between 0 & 1, & multiply it by the height of the rectangle of ABCD.

$f(x) = x^3$

Now check if $y \le x^3$; If so the point lies below the curve. have $y \le x^3$ is added to M. N = no. of trials.

→ কীভাবে বুঝবো (x, y) curve-এর নিচে না উপরে,

if $y \le f(x)$ than point curve-এর নিচে

if $y > f(x)$ " " " উপরে.

R/N	x $0.1R$	R/N(R)	$y = 140 \times R$	x^3	M	N
22	2.2	.57	79.8	10.65	0	1
25	2.5	.18	25.2	15.63	0	2
27	2.7	0.00	0.0	19.68	1	3
45	4.5	.90	126.0	91.13	0	4
48	4.8	.42	58.8	110.82	2	5
40	4.0	.77	107.8	64.00	0	6
⋮		⋮	⋮			7
						8
					8	20

$y \le x^3$

$$\therefore I = \frac{8}{20} \times A = \frac{8}{20} \times 420 = 168.00$$

Actual value of Integral = 152.25.

By increasing the no. of points value of I quite close to the actual value. can be obtained & then larger number of points can be generated by a

(2017-18) Define Monte carlo method. Solve the integral $I = \int_2^5 x^3 dx$ by monte carlo ... [Use Random Number 22, 25, 18, 45, 15, 27, 48, 43, 40, 47 for x] & (.57, .18, .00, .90, 0.05, 0.77, .66, .10, .76, .42) for y.

(Example-2.9) Determination of value of π by Monte Carlo. $\pi = 3.1416$.

→ The value of π can be determined by using the relation for area of a circle

∴ Area of a circle $= \pi r^2$.

Area of a ~~quard~~ quadrant $= \frac{\pi r^2}{4} = \frac{\pi}{4}$.

$r = 1$

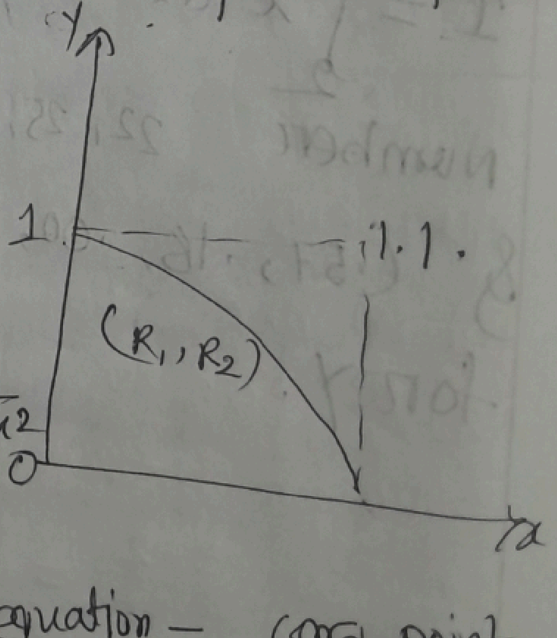
→ Quadrant -এর equation

$= x^2 + y^2 = 1$

If In → $x^2 + y^2 \leq 1$.

$y^2 \leq 1 - x^2 \Rightarrow y \leq \sqrt{1-x^2}$

out ⇒ $x^2 + y^2 > 1$.



→ $x^2 + y^2 = 1$ → ? circle-এর equation — কোন point (x, y) যদি origin থেকে Distance r হয়, তাহলে,

$$Dist = \sqrt{x^2 + y^2}$$

$x = R_1$ $\quad x^2+y^2 \le 1.$

$y = R_2$ $\quad \Rightarrow y^2 \le 1-x^2 \Rightarrow y \le \sqrt{1-x^2}$

circle = যেসব point-এর origin থেকে Dist equal r.

$$\sqrt{x^2+y^2} = r.$$

$$\Rightarrow x^2+y^2 = r^2. \qquad r=1.$$

$$\therefore x^2+y^2 = 1$$

→ quadrant থাকে square-এর মধ্যে, $A_{square} = 1\times1 = 1$

R_1 = random x (0 থেকে 1)

R_2 = random y (0 থেকে 1).

N → $R_1(x)$	$R_2(y)$	$R_2 \sqrt{1-R_1^2}$ x^2+y^2	IN/out	N.
0.82	0.95	.37124	out	1
.34	.14	.9408.	IN	2
.20	.84	0.9797	IN	3
.58	.14.	0.8146	IN.	4
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
			IN.	40

M = 31. N = 40

$$\frac{\pi}{4} = \frac{M}{N} \times A = \frac{31}{40} \times 1$$

$$\Rightarrow \pi = \frac{31}{40} \times 4 = 3.10.$$

why ? N বাড়ালে accuracy বাড়ে -

- Random fluctuation কমে।
- Monte carlo Law Large number-এ কাজ করে।

The accuracy of the result can be increased by increasing the number of points (N).

(2.5) Random walk problem →. A drunkard moves.

Using single digit random numbers. →.

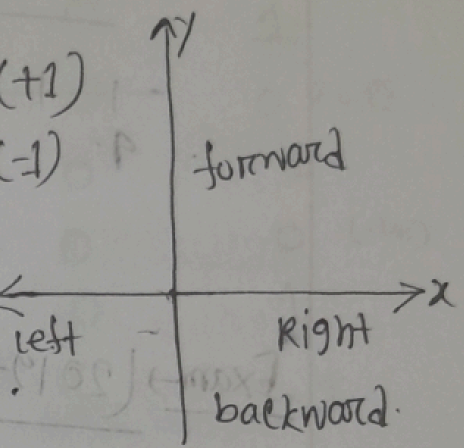
***

→ forward এ গেলে y-axis এ 1 বাড়বে (+1)

Backward - - - y - - - 1 কমবে (-1)

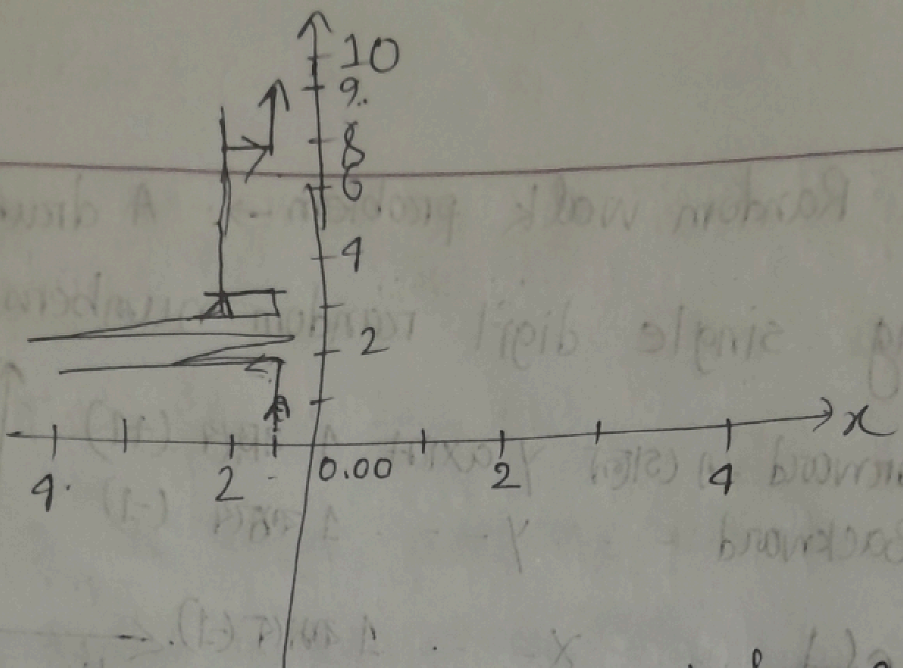
→ Left - - x - - 1 কমবে (-1).

Right - x - - 1 বাড়বে (+1).



Direction	Given Proba.	R/N
Forward (F):	0.5 (50%)	0,1,2,3,4
Left (L);	0.3 (30%)	5, 6, 7
Right (R):	0.2 (20%)	8, 9

Step	R/N	Dir	Pos x	Pos y
1	6	L	-1	0
2	2	F	-1	1
3	0	F	-1	2
4	0			
5	6	L	-2	2
6	8	R	(-2+1) -1	2
	5	L	-2	2
20	4			
		F	-1	9



Exam → (2019-20), A drukard moved from a point to a destination & takes step in four direction, backward, forward, right, left.

Given,

Dire	Prob	R/N
Forward (F)	- 50% - 0.5	31 - 80 cm
Backward (B)	- 10% - 0.1	1 - 30 cm
Left (L)	- ~~25%~~ 25% - 0.25	31 - 40 cm
Right (R)	- 15% - 0.15	41 - 60 cm

→ (back, for, Right, Left - 30 cm, 80 cm, 60 cm, & 40 cm)

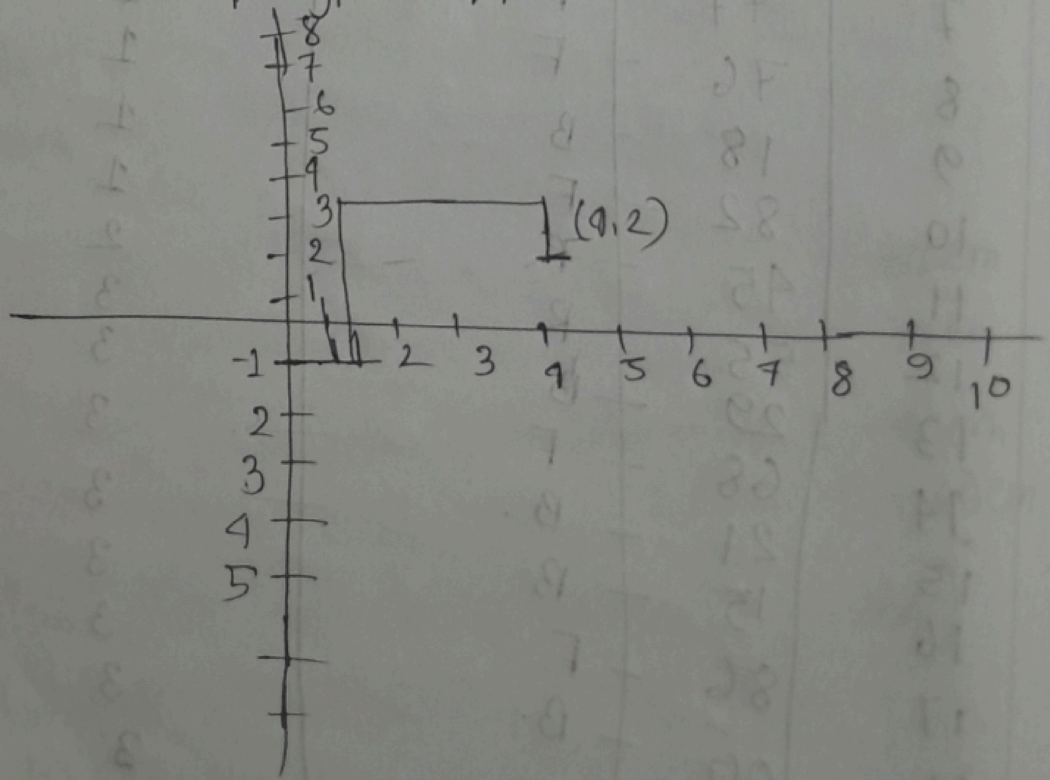
forward → y++
back → y--
Right → x++
left → x--

1-1

end of 20 steps.

step	R/N	Direction	Position x	Position y
1	67	F	0	1
2	23	B	0	0 (1-1)
3	01	B	0	-1
4	95	F	0	0 (-1+1)
5	81	F	0	1
6	56	R	1	1
7	77	F	1	2
8	76	F	1	3
9	18	B	1	2 (3-1)
10	82	F	1	3
11	45	R	2	3.
12	55	R	3	3
13	29	B	3	2
14	68	F	3	3
15	21	B.	3	2
16	15	B	3	1
17	86	F	3	2
18	00	B.	3	1
19	89	F	3	2
20	43	R.	4	2

when the drunkard takes steps in forward direction, the value of y increased by 1. If drunkard takes steps in backward the value of y decreased by 1. If drunkard takes steps in left of x decreased by 1. & for Right x increased " 1. The simulation for 20 steps is given in table.



Normally Distributed Random Numbers →

→ In many real life systems, the outcome do not spread evenly, instead, they cluster around an average (mean).

<u>Example:</u> marks obtained by student.

Measurement errors.

Bomb falling around an aiming point

→ A normal distribution Parameters →.

- mean (μ) → center value.
- Standard deviation (σ) → spread of data.

A normal random variable is generated using.

$$x = \mu + \sigma z$$

x = generated value.

μ = mean.

σ = Standard deviation.

z = Standard normal random number (mean = 0, std = 1)

$$y = \mu + \sigma z$$

z = random normal number

Why we Random Normal Number in Simula

- uniform random numbers (0-1) are not enough when behavior is normal.
  → We use (12 uniform-number-method)
-

→ (Ex-2.7) Application of RNN.

Target Area ; Rectangular
→ X-direction = 1000 m.
Y-direction = 600 m.
- center of target = (0,0).
- Bomb aim at the center.
- Bomb impacts points are normally distributed.

Standard deviation → $\sigma x = 500$ m.

$\sigma y = 300$m.

- A bomb is hit, if it lands inside the ~~rectangule~~ rectangle.

$X = 500\, z_x$, $\quad$ z_x, z_y → normal Random number.

$Y = 300\, z_y$. $\quad$ Mean $= 0$.

x range $= -500$ to $+500$

y range $= -300$ to $+300$.

Bomb strike	RNN(z_x)	$x = 500 \times$ RNN(x)	RNN(y)	$y = 300 \times$ RNN(y)	Rs.
1	.23	115	.24	72	Hit
2	-1.16	-580	-0.02	-6	miss
3	0.39	195	0.64	192	Hit
4	-1.90	-950	~~312~~ -1.04	312	miss
5	-0.78	-396	~~2.04~~ 0.68	2.04	Hit
6	-0.02	-10	~~141~~ -0.47	141	"
7	-0.40	-200	~~-225~~ -0.75	-225	"
8	-0.66	-330	~~-132~~ -0.44	-132	Hit
9	1.41	-705	~~363~~ 1.21	363	miss
10	0.07	35	~~-24~~ -0.08	-24	Hit
20	-0.21	-105	0.65	195	Hit

Number of Hits $= 10$

" " Miss $= 10$

% Number of strikes on target $= 50\%$.

Book (Ex-6) Exercise

$x_1 = 500 z x$ $\quad x^2+y^2 \leq R^2$

$y_1 = 300 z y$. $\quad x^2+y^2 \leq R^2$

$R = 500m$. To hit.

No. of	RNN	x_1	RNN	y_1	$x^2+y^2 \leq R^2$	Result.
1	.23	115	.24	72		
2	-1.16	-580	-0.02	-6		
3.	0.39	195	0.64	192		
4	-1.90	-950	-1.04	-312		
5	-0.78	-390	0.68	204		
6	-0.02	-10	-0.47	-141		
7	-0.40	-200	-0.75	-225		
8	-0.66	-330	-0.44	-132		
9	1.41	705	1.21	363		
10	0.07	35	-0.08	-24		
11	0.53	265	-1.56	-468		
12	0.03	15	1.49	447		
13	-1.19	-595	-1.19.	-357.		

Exercise (1) Determine Area of irregular figure -

Here, $\frac{F}{A} \doteq \frac{M}{N}$

F = Area of irregular
A = Area of square to be considered.
M = Points within shape.
N = Total random points.

Random number between (0-1)

For all cases we will consider 9×9 cm² regular shape.

-) Random Number table:

$x_1 = 0.96 \times 9 = 8.64$
$x_2 = 0.71 \times 9 = 6.39$
$x_3 = 0.03 \times 9 = 0.27$

$y_1 = 0.42 \times 9 = 3.78$
$y_2 = 0.10 \times 9 = 0.9$
$y_3 = 0.30 \times 9 = 2.7$

Area, $F = \frac{26}{84} \times (9 \times 9) = 25.07$ cm².

considering ration of points falling $\frac{M}{N} = 30.95\%$
$= \frac{26}{84}$.